

SEO & GEO Parameter Reference (Automation Plan)

Complete parameter checklist for Search Engine Optimisation (SEO) and Generative Engine Optimisation (GEO), showing for each parameter: what to check, how the tool checks it, whether it is Automated or Manual, and the API key / tool planned.

Policy:

- SEO is crawl-based. Parameters that cannot be determined from the site itself are marked Manual and excluded from the SEO score.
- **Connected mode:** six first-party SEO parameters (CTR, sitemap GSC submission, log/crawl analysis, dwell time, bounce rate, pogo-sticking) become **Measured** when the domain is in ``backend/connected_properties.json`` and OAuth or service-account credentials have API access. If connected but a specific API returns no rows for the page, that parameter stays Manual until data exists.
- GEO is fully automated using external APIs plus Gemini. GEO has no Manual parameters.

Detection method legend:

- Crawl = httpx + Playwright/Firecrawl + BeautifulSoup parsing of the page, headers, robots/sitemap/lms.txt
- PSI = Google PageSpeed Insights API (Lighthouse + lab Core Web Vitals + CrUX field data)
- LLM = Gemini judgement
- External = third-party API (GEO only)
- First-party = GA4 / Search Console via OAuth token or service account (connected domains only)
- Manual = requires data the tool cannot obtain (site ownership / off-site data); shown but not scored

API keys referenced (from project .env):

- GEMINI_API_KEY (present), OPENAI_API_KEY (present), GOOGLE_API_KEY (present - PageSpeed + Knowledge Graph)
- SERPER_API_KEY (present), PERPLEXITY_API_KEY (present), FIRECRAWL_API_KEY (present), X_BEARER_TOKEN (present)
- SERPAPI_API_KEY (present - Google AI Overview citations)
- OAuth (connected mode): ``secrets/google-oauth-client.json`` + ``secrets/google-oauth-token.json`` (optional ``GOOGLE_OAUTH_*`` in ``.env``)
- Service account (optional fallback): ``GOOGLE_APPLICATION_CREDENTIALS``
- Wikidata SPARQL and Wikipedia REST API need no key

Part 1 - SEO Parameters

01 Crawlability & Indexability

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
robots.txt	Correct disallow rules; no accidental crawl blocks on key pages	Fetch and parse /robots.txt	Automated	Crawl (no key)
sitemap.xml	Present; no 4xx/5xx URLs; lastmod populated	Fetch sitemap, validate URLs + status	Automated	Crawl (no key)
sitemap GSC-submission status	Submitted to Google Search Console	GSC sitemaps.list/get (connected), else Manual	Connected (GSC)	OAuth token or GOOGLE_APPLICATION_CREDENTIALS
Meta robots	No unintended noindex/nofollow; check X-Robots-Tag header	Parse meta tag + response headers	Automated	Crawl (no key)
Canonical tags	Self-referencing; no conflicting/chained canonicals	Parse <link rel=canonical>	Automated	Crawl (no key)
HTTP status & redirects	No redirect chains > 1 hop; 301 not 302 for permanent	Follow + record redirect chain	Automated	Crawl (no key)
Crawl depth (NEW)	Key pages reachable within 3 clicks from homepage	Limited internal crawl + depth graph	Automated	Crawl (no key)
Pagination handling (NEW)	rel=next/prev or infinite-scroll pattern; no thin dupes	Parse pagination markup	Automated	Crawl (no key)
Log file analysis (NEW)	Googlebot visit frequency; crawl budget waste	GSC urlInspection (connected), else Manual	Connected (GSC)	OAuth token or GOOGLE_APPLICATION_CREDENTIALS
XML sitemap index (NEW)	Sitemap index pointing to sub-sitemaps (news/image/video)	Detect <sitemapindex>	Automated	Crawl (no key)

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
noarchive / nosnippet (NEW)	No accidental suppression of snippets or cached pages	Parse robots directives + headers	Automated	Crawl (no key)

02 On-Page Signals

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Title tag	50-60 chars, primary keyword near front, unique	Parse <title>, length + keyword position	Automated	Crawl (no key)
Meta description	150-160 chars, action-oriented, includes target keyword	Parse meta description	Automated	Crawl (no key)
Heading hierarchy	Single H1; logical H2->H3; no skipped levels	Parse heading tree	Automated	Crawl (no key)
Word count / content depth	Matches competitive benchmark; no thin content	Count words + Gemini depth judgement	Automated	Crawl + GEMINI_API_KEY
Image alt coverage	Informational images have alts; decorative use alt=""	Parse coverage	Automated	Crawl (no key)
Internal links + anchor text	Keyword-rich anchors; no orphan pages; logical graph	Parse links + build internal graph	Automated	Crawl (no key)
Keyword usage	Primary keyword in title/H1/first 100 words; related terms	Text analysis + Gemini semantic check	Automated	Crawl + GEMINI_API_KEY
Duplicate content (NEW)	Near-duplicate pages consolidated via canonical/301	Compare page fingerprints + canonicals	Automated	Crawl (no key)
Content freshness signals (NEW)	Visible publish + last-updated dates	Parse dates in HTML + schema	Automated	Crawl (no key)
Table of contents (NEW)	Anchor-linked ToC on long-form pages	Detect ToC anchors	Automated	Crawl (no key)
E-E-A-T signals on-page (NEW)	Author bio, credentials, About page, attribution	Parse author/about + Gemini judgement	Automated	Crawl + GEMINI_API_KEY
Outbound link quality (NEW)	Links to authoritative, topically relevant external sources	Parse outbound links + Gemini relevance	Automated	Crawl + GEMINI_API_KEY
Multimedia coverage (NEW)	Video, images, or infographics present	Detect media elements	Automated	Crawl (no key)
LSI / semantic coverage (NEW)	Topic completeness vs top-ranking pages	Gemini topic-model comparison	Automated	GEMINI_API_KEY

03 Technical SEO

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
HTTPS / SSL	Valid cert; no mixed content; HSTS header	Inspect TLS + headers + page resources	Automated	Crawl (no key)
Mobile viewport	Meta viewport; no horizontal scroll; tap targets >= 44px	Parse viewport + PSI mobile audit	Automated	Crawl + PSI (GOOGLE_API_KEY)
URL structure	Lowercase, hyphens, keyword-rich, < 115 chars, no junk params	Analyze URL string	Automated	Crawl (no key)
hreflang	Correct lang/region; self-referencing; x-default set	Parse hreflang tags	Automated	Crawl (no key)
Broken links	0 internal 4xx; external link monitoring	Crawl links + check status codes	Automated	Crawl (no key)
Render-blocking resources	CSS/JS deferred or async; critical CSS inlined	PSI audit + markup parse	Automated	PSI (GOOGLE_API_KEY)
JavaScript rendering (NEW)	Critical content not JS-only	Diff raw HTML vs rendered DOM	Automated	Crawl (Playwright/Firecrawl)
HTTP/2 or HTTP/3 (NEW)	Protocol version check	Inspect connection protocol	Automated	Crawl (no key)
Server response time (NEW)	TTFB < 200ms; server-side caching	Measure TTFB + PSI	Automated	Crawl + PSI (GOOGLE_API_KEY)
Image compression & format (NEW)	WebP/AVIF; lazy loading; explicit width/height	PSI image audit + markup parse	Automated	PSI (GOOGLE_API_KEY)
Third-party script impact (NEW)	Defer non-critical scripts	PSI third-party audit	Automated	PSI (GOOGLE_API_KEY)
Security headers (NEW)	CSP, X-Frame-Options, Referrer-Policy	Inspect response headers	Automated	Crawl (no key)
AMP / Web Stories (NEW)	Valid AMP markup + AMP-canonical pairing	Detect AMP markup	Automated	Crawl (no key)

04 Core Web Vitals & Performance

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
LCP	<= 2.5s; main image/text block is largest element	PSI lab + CrUX field	Automated	PSI (GOOGLE_API_KEY)
CLS	<= 0.1; reserve space for media/embeds	PSI lab + CrUX field	Automated	PSI (GOOGLE_API_KEY)
INP	<= 200ms; replaces FID	PSI / CrUX field	Automated	PSI (GOOGLE_API_KEY)

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
TTFB	<= 800ms; server + CDN tuning	PSI / measured	Automated	PSI (GOOGLE_API_KEY)
Lighthouse scores	Performance, Accessibility, Best Practices, SEO >= 90	PSI category scores	Automated	PSI (GOOGLE_API_KEY)
Field data (CrUX) (NEW)	Real-user field data; lab != real users	PSI CrUX data	Automated	PSI (GOOGLE_API_KEY)
Mobile vs desktop delta (NEW)	Run CWV separately for mobile and desktop	PSI run twice	Automated	PSI (GOOGLE_API_KEY)
Font loading strategy (NEW)	font-display: swap; preload key fonts; fallbacks	Parse CSS/font usage + PSI	Automated	Crawl + PSI (GOOGLE_API_KEY)

05 Structured Data & Social

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Schema.org JSON-LD	Article, Product, FAQ, HowTo, BreadcrumbList, Organization, LocalBusiness	Parse + validate JSON-LD	Automated	Crawl (no key)
OG / Twitter cards	og:title, og:image (1200x630), og:description, twitter:card	Parse meta tags	Automated	Crawl (no key)
Favicon	SVG + 32px PNG + apple-touch-icon	Detect favicon links	Automated	Crawl (no key)
Rich result eligibility (NEW)	Warnings/errors check	Validate schema against rich-result rules	Automated	Crawl (no key)
Breadcrumb schema (NEW)	BreadcrumbList on interior pages	Parse JSON-LD	Automated	Crawl (no key)
Sitelinks searchbox (NEW)	WebSite schema with potentialAction SearchAction	Parse JSON-LD	Automated	Crawl (no key)
Review / Rating schema (NEW)	AggregateRating on product/service pages	Parse JSON-LD	Automated	Crawl (no key)
Video schema (NEW)	VideoObject with thumbnail, duration, uploadDate	Parse JSON-LD	Automated	Crawl (no key)

06 Off-Page & Authority

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Backlink profile	Domain authority distribution; toxic ratio; anchor diversity	Requires off-site link index	Manual	Requires off-site link data
Referring domain count	Unique linking root domains; trend	Requires off-site link index	Manual	Requires off-site link data
Topical authority	Topic cluster breadth; hub-spoke internal linking	Internal graph + Gemini topic analysis	Automated	Crawl + GEMINI_API_KEY
Brand mentions (unlinked)	Citation signals across the web	Requires off-site monitoring	Manual	Requires off-site monitoring
Competitor link gap	Links competitors have that you lack	Requires off-site link index	Manual	Requires off-site link data
NAP consistency (local) - on-page	Name/Address/Phone present and consistent on page	Parse on-page NAP	Automated	Crawl (no key)
NAP consistency (local) - cross-directory	Identical across external directories	Requires directory data	Manual	Requires off-site data
Google Business Profile	Verified, complete listing; reviews; posts	Requires GBP ownership/listing data	Manual	Requires GBP

07 UX & Engagement Signals

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Click-through rate (CTR)	GSC CTR vs position benchmark	GSC searchanalytics.query (connected + data), else Manual	Connected (GSC)	OAuth token or GOOGLE_APPLICATION_CREDENTIALS
Dwell time / scroll depth	GA4 engagement proxy	GA4 runReport (connected + data), else Manual	Connected (GA4)	OAuth token or GOOGLE_APPLICATION_CREDENTIALS
Bounce rate in context	High bounce interpreted by page type	GA4 runReport bounceRate (connected + data), else Manual	Connected (GA4)	OAuth token or GOOGLE_APPLICATION_CREDENTIALS
Pogo-sticking risk	Users returning to SERP quickly = intent mismatch	GA4 + GSC derived (connected + data), else Manual	Connected (GA4+GSC)	OAuth token or GOOGLE_APPLICATION_CREDENTIALS

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Search intent alignment	Informational/navigational/commercial/transactional match	Gemini judges intent vs page type	Automated	GEMINI_API_KEY

Part 2 - GEO Parameters (fully automated)

01 Answerability

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Concise top answer	Direct answer in first 40-50 words	Parse intro + Gemini judgement	Automated	Crawl + GEMINI_API_KEY
Question-phrased headings	H2/H3 as natural-language questions	Parse headings	Automated	Crawl (no key)
FAQ section	Marked up with FAQPage schema; long-tail variants	Parse JSON-LD + content	Automated	Crawl (no key)
TL;DR / summary box	Scannable summary easy for AI to lift	Detect summary + Gemini	Automated	Crawl + GEMINI_API_KEY
Definition blocks (NEW)	"X is defined as..." near the top	Pattern + Gemini detection	Automated	Crawl + GEMINI_API_KEY
Step-by-step answers (NEW)	Numbered steps with clear action verbs	Parse ordered lists	Automated	Crawl (no key)
Comparison answers (NEW)	Tables or "A vs B" sections for comparative queries	Detect comparison tables	Automated	Crawl (no key)

02 Extractability

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Semantic HTML	article/section/aside/figure used correctly; not div-soup	Parse DOM structure	Automated	Crawl (no key)
Chunkable paragraphs	Short paragraphs (2-4 sentences); one idea each	Analyze paragraph lengths	Automated	Crawl (no key)
Lists & tables	HTML ul/ol/table (not image-based)	Detect list/table elements	Automated	Crawl (no key)
Heading-to-content ratio (NEW)	Each heading followed by substantive content	Map headings to content blocks	Automated	Crawl (no key)
Code blocks / data blocks (NEW)	pre/code tags; machine-readable tables	Detect code/data markup	Automated	Crawl (no key)
No content in images/PDFs (NEW)	Text not locked inside images/PDFs	Detect text-in-image + Gemini OCR check	Automated	Crawl + GEMINI_API_KEY
Consistent terminology (NEW)	Same term used throughout; no synonym-switching	Gemini terminology analysis	Automated	GEMINI_API_KEY

03 Citability & E-E-A-T

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Author + credentials	Named author, bio, qualifications visible	Parse author block + Gemini	Automated	Crawl + GEMINI_API_KEY
Sources & citations	Outbound links to primary sources; references section	Parse citations/links	Automated	Crawl (no key)
Statistics & data	Quantitative claims with source attribution	Detect stats + Gemini	Automated	Crawl + GEMINI_API_KEY
Publish + updated dates	ISO 8601 dates in HTML + Article schema	Parse dates + JSON-LD	Automated	Crawl (no key)
About / Trust pages (NEW)	About, Contact, Privacy pages present	Fetch + detect trust pages	Automated	Crawl (no key)
Expert review signals (NEW)	"Medically reviewed by..."/"Fact-checked by..." labels	Pattern + Gemini detection	Automated	Crawl + GEMINI_API_KEY
Wikidata / Wikipedia entity (NEW)	Organisation/person entity exists	Query Wikidata + Wikipedia	Automated	Wikidata SPARQL + Wikipedia (no key)
Press mentions & awards (NEW)	Featured-in logos and award badges	On-page detection + Serper web search	Automated	Crawl + SERPER_API_KEY
Disclaimer / methodology (NEW)	Methodology explained for research/data content	Detect methodology + Gemini	Automated	Crawl + GEMINI_API_KEY

04 AI Structured Data

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
FAQPage schema	Q&A pairs in JSON-LD	Parse JSON-LD	Automated	Crawl (no key)

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
HowTo schema	Steps with name, text, image	Parse JSON-LD	Automated	Crawl (no key)
Article schema	headline, author, datePublished, dateModified, publisher	Parse JSON-LD	Automated	Crawl (no key)
Organization schema	name, url, logo, sameAs (socials, Wikidata)	Parse JSON-LD	Automated	Crawl (no key)
Product / Offer schema	name, description, price, availability, aggregateRating	Parse JSON-LD	Automated	Crawl (no key)
SpeakableSpecification (NEW)	Important passages marked for voice/AI	Parse JSON-LD	Automated	Crawl (no key)
Dataset schema (NEW)	For research/data pages	Parse JSON-LD	Automated	Crawl (no key)
Event schema (NEW)	For time-bound content	Parse JSON-LD	Automated	Crawl (no key)
ClaimReview schema (NEW)	Fact-check pages	Parse JSON-LD	Automated	Crawl (no key)

05 Entity Clarity

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
In-text definitions	First use of technical terms defined inline / linked	Detect definitions + Gemini	Automated	Crawl + GEMINI_API_KEY
Glossary page	Dedicated glossary with anchor links	Detect glossary page	Automated	Crawl (no key)
Named entities marked up	People/places/orgs in schema with sameAs to authoritative URIs	Parse JSON-LD sameAs	Automated	Crawl (no key)
Disambiguation (NEW)	Clarify ambiguous terms near first use	Gemini ambiguity analysis	Automated	GEMINI_API_KEY
Knowledge panel consistency (NEW)	Brand name/description/logo match Knowledge Panel	Query Google Knowledge Graph	Automated	GOOGLE_API_KEY
Entity co-occurrence (NEW)	Mention related authoritative entities	Gemini entity analysis	Automated	GEMINI_API_KEY

06 AI-Bot Crawlability

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
GPTBot / ClaudeBot / PerplexityBot	Explicit allow/disallow in robots.txt	Parse robots.txt	Automated	Crawl (no key)
Google-Extended	Controls AI Overviews training data	Parse robots.txt	Automated	Crawl (no key)
llms.txt	Machine-readable site summary for LLMs	Fetch /llms.txt	Automated	Crawl (no key)
CCBot (Common Crawl) (NEW)	Explicit allow/disallow	Parse robots.txt	Automated	Crawl (no key)
Applebot-Extended (NEW)	Apple Intelligence crawler directive	Parse robots.txt	Automated	Crawl (no key)
Meta AI crawlers (meta-externalagent) (NEW)	Meta AI training/retrieval directive	Parse robots.txt	Automated	Crawl (no key)
Crawl rate limits (NEW)	AI bots not throttled to stale content	Parse Crawl-delay + measure response	Automated	Crawl (no key)
llms-full.txt (NEW)	Extended page-level metadata file	Fetch /llms-full.txt	Automated	Crawl (no key)

07 AI Quality Scores

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Clarity score	Writing unambiguous and scannable?	Gemini judgement	Automated	GEMINI_API_KEY
Authority score	Expertise, credentials, sourcing signaled?	Gemini judgement	Automated	GEMINI_API_KEY
Comprehensiveness	Covers topic fully enough to be self-sufficient?	Gemini judgement	Automated	GEMINI_API_KEY
Citation likelihood	Would an AI engine prefer this source?	Gemini judgement	Automated	GEMINI_API_KEY
Sample-query self-sufficiency test	Ask 5-10 likely queries; does page alone answer each?	Gemini multi-query test	Automated	GEMINI_API_KEY
AI Overview appearance monitoring (NEW)	Cited in Google AI Overviews / SearchGPT / Perplexity?	Run queries; check citations	Automated	SERPAPI_API_KEY + PERPLEXITY_API_KEY + OPENAI_API_KEY
Hallucination surface area (NEW)	Reduce ambiguous claims AI might misattribute	Gemini ambiguity/scope analysis	Automated	GEMINI_API_KEY

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Prompt-align test (NEW)	Simulate LLM summary; accurate and favourable?	Gemini summary simulation	Automated	GEMINI_API_KEY
Competing source gap analysis (NEW)	Which sources AI cites for key queries that you lack	Diff citation lists vs target domain	Automated	SERPAPI_API_KEY + PERPLEXITY_API_KEY

08 Multimodal & Voice GEO

Parameter	What to check	How it's checked	Auto/Manual	API key / tool
Image alt text quality	Descriptive alts that match image content	Parse alts + Gemini vision check	Automated	Crawl + GEMINI_API_KEY
Video transcripts	Full text transcripts on-page or in schema	Detect transcript/VideoObject	Automated	Crawl (no key)
Conversational phrasing	Some content in natural spoken language for voice	Gemini tone analysis	Automated	GEMINI_API_KEY
Featured snippet shape	Match Google's voice/AI format: paragraph, list, or table	Detect content shape + Gemini	Automated	Crawl + GEMINI_API_KEY
SpeakableSpecification (voice)	Passages marked for text-to-speech	Parse JSON-LD	Automated	Crawl (no key)

Summary

- SEO: 11 parameters are Manual when a domain is ****not connected**** (excluded from score): Log file analysis, sitemap GSC-submission status, Backlink profile, Referring domain count, Brand mentions, Competitor link gap, NAP cross-directory, Google Business Profile, CTR, Dwell time/scroll depth, Bounce rate, Pogo-sticking. When connected via OAuth/service account, ****6** of these become Measured****** (Manual count drops to 5 off-page/GBP params).
- GEO: 0 Manual parameters - fully automated via Crawl + Gemini + External APIs.
- Connected mode setup is documented separately: ``Connected_Mode_Site_Owner_Guide.pdf`` and ``Connected_Mode_Tool_Operator_Guide.pdf`` (regenerate from markdown via ``generate_connected_guide_pdf.py``).